

Version 9 Accuracy Program

Full Status Report

Steel Beam Estimator — SteelBeamEstimator V9

Date: 2026-08-04 | Current MODEL_VERSION: 9.3.4 | Baseline: V8.9.5

BASELINE	CURRENT	TARGET	GAP
Overall: 58.07%	70.23%	≥80 / 80 / 72%	~10 pts
Bar detection: 41.52%	68.52%	≥85%	~16.5 pts
Bar match: 24.16%	27.41%	≥70%	~42.6 pts
Steel KG: 72.69%	91.07%	≥92%	~1 pt

1. Original Plan vs Reality — What Changed and Why

The original plan had three tracks running sequentially over ~8 weeks. Reality diverged in two important ways — both positive surprises.

Surprise 1 — Week 1: Discovery gap, not association gap

The Day-1 stirrup diagnostic found that 90.4% of GT stirrup rows were simply never discovered as text. Set 1 was 0%. This meant the original Track 1 premise ('annotations exist but never associated') was **wrong for the dominant volume**. The fix was a one-line entity-type addition to `_collect_entities` (the 9.2.0 DIMENSION-channel patch), not a multi-week geometry engine. This single patch recovered more ground than Track 1 was originally budgeted to recover.

Surprise 2 — Render/crop infrastructure complexity (ongoing)

Track 1's crop/render infrastructure, which the original plan assumed was ready (promoted from M.1), turned out to have a compounding set of defects: full-sheet rendering at wrong resolution, coarse padding bleeding into neighbors, and a structural gap where R.1 text anchors don't cover the tick-row Y-band for ~40% of residual beams. This has absorbed significant iteration and is still not fully resolved.

Plan Changes Summary

Original Plan	What Actually Happened	Reason
Track 1 was primary stirrup recovery	9.2.0 DIMENSION patch did most of the work	Day-1 diagnostic found discovery gap
T1.1 renderer assumed ready from M.1	Required 5+ iterations of fixes	M.1 built for datasets, not detection
Track 1 effort: ~4 weeks	Ongoing past estimate	Hidden renderer infrastructure complexity
Track 2 scoped for weeks 4–8	Not started — blocked on crop quality	Cannot send blank crop to VLM
T0.2a hooks in week 2	Not started	Deprioritised pending stirrup stabilisation
T0.3 face gate before Track 2	Not started	Waiting for Track 1 crop work to settle
Vertical crop from geometric signals	Never implemented — always R.1-anchor-derived	Spec didn't specify this explicitly enough
OpenCV fallback expected to work	Was never running (cv2 not installed until 9.3.2)	Missing from requirements.txt

2. What Has Been Achieved

2.1 Phase / Version Timeline

Version	Phase	Status	Key Output
9.1.0	M.2 — Spacer Rule Engine (T0.1)	Committed	Spacer Missing 98→33
9.2.0	DIMENSION-channel discovery patch	Committed	Stirrup Missing 182→73, +7.2% overall
9.3.0	Track 1 — Geometric evidence engine	Committed	Stirrup Missing 73→31, -57%
9.3.1	Track 1 Closure — R4/R6/R3 proofs	Committed	Propagation fix, flag-off clean
9.3.2	OpenCV activation (dependency fix)	Committed	cv2 installed, 0 accuracy improvement
9.3.3	Crop mechanism fix (local-extent render)	Committed	B1/B2 correct; B8/B9/B10 still wrong
9.3.4	Adjacent-beam split + Y-axis diagnosis	In Progress	Horizontal split fixed; Y-axis gap exposed

2.2 Accuracy Journey Across All 3 Sets

Metric	V8 Baseline	After 9.1.0	After 9.2.0	After 9.3.1	Target
Overall %	58.07	60.79	67.97	70.23	≥80/80/72
Bar detection %	41.52	48.99	61.37	68.52	≥85
Bar match %	24.16	25.05	25.04	27.41	≥70
Steel KG %	72.69	75.20	91.57	91.07	≥92
STIRRUP Missing	182	—	73	31	~0
SPACER Missing	98	33	33	33	~0

Note: Steel KG is the metric closest to target (91% vs ≥92%). Bar match at 27% is the largest gap — the entire motivation for Track 2.

2.3 What Each Phase Actually Did

M.2 / 9.1.0 — Spacer Rule Engine (T0.1)

Implemented the hardcoded Ø25@1000mm estimation-team rule. Correctly closed 65 of 98 missing spacer rows. The residual 33 are structurally upstream-dependent — they require a MAIN+EXTRA pair to exist in the model first, which Track 2 (extra-bar recovery) will supply. The QA.2A ACCEPTABLE_EXTRA carve-out was shipped alongside so GT-omission spacers are not penalised. Parked at 33 with a documented tracking list of 19 confirmed absent-pair beams. Constants SPACER_DIA_MM=25 and SPACER_SPACING_MM=1000 are hardcoded non-negotiable. Cover sourced from R.2A EngineeringContext (cover_beam_mm). All 15 unit-test vectors green.

9.2.0 — DIMENSION-channel discovery patch

The single most impactful change in the entire program. Patch: 7 files, one entity-type addition (DIMENSION text overrides to _collect_entities), plus a runner path fix (V9 runner was silently loading V8 source — a pre-existing bug this patch incidentally fixed). Discovery ratios: Set1 0%→104%, Set2 3%→71%, Set3 17%→72%. Also confirmed V8/V9 R.1 source was byte-identical pre-patch, so all prior baselines remain valid. R4 flag-off: IDENTICAL on all 3 sets. R5: zero non-STIRRUP production deltas.

9.3.0 / 9.3.1 — Track 1 Geometry Engine

Built T1.1 (renderer validation gate — all 4 checks PASS), T1.2 (vector tick detection + OpenCV fallback), T1.3 (GEOMETRY_STIRRUP fusion in R.2.1D: AGREE/CONFLICT/TEXT_ONLY/GEOMETRY_ONLY_SYNTH), T1.4 (Type3 zone boundary refinement from pitch-change measurement). Cut Stirrup Missing from 73→31 (–57%). WRONG_QTY 100→113 explained as 72% already-broken beams redone differently (Category A), 27% previously-MISSING beams now emitting nonzero quantities the taxonomy calls WRONG_QTY but which represent genuine progress. Closure passed R4 (35/35), R6 (144/144), R3 (confidence propagation fixed for AGREE/CONFLICT rows). The 2 pure-geometry-only beams (B35A, B25A) get advisory notes only.

9.3.2 — OpenCV Activation

Installed opencv-python-headless==5.0.0.93, fixed HoughLinesP shape mismatch for OpenCV 5. Result: 7 of 74 previously-dead beams got real OpenCV detections, but all 7 had confidence $0.45 < \text{fusion bar } 0.55$ — zero production change. Confirmed the residual is NOT an environment gap but genuine signal weakness for these specific beams. requirements.txt updated with pinned version.

9.3.3 — Crop Mechanism Fix

Replaced full-sheet-render-then-slice (producing ~100-130px crops at wrong resolution) with local-extent rendering at fixed 1200px longer side. Fixed horizontal neighbor-splitting for packed rows using hard non-overlap boundary at label midpoints. B1/B2 crops now correct and match manual references. B8/B9/B10 improved but still wrong — exposing the Y-axis structural gap now under investigation in 9.3.4.

3. What Is Yet to Be Achieved

3.1 Immediate — Still in 9.3.4

Item	Status	Blocker / Note
Y-axis crop extent structural fix (geometric floor)	In Progress — approved, not yet coded	Pivot confirmed by scaled diagnostic (40% Pattern A)
Part C content-based visual gate (4-check)	Blocked	Cannot gate on text/label alone — tick geometry must be visibly present
Full 115-beam Part D re-run + QA.2A delta	Blocked	Depends on Part C pass
B8 multi-zone render legibility	Not started	Legitimately fragmented beam — needs visual note, not coverage fix
B4/B5/B7 stale-artefact purge	Minor cleanup	Not blocking

3.2 Near-term After 9.3.4

Item	Phase	Notes
T0.2a — Stirrup hook generation	Separate phase	81 STIRRUP_HOOK missing. Hook length from R.2A anchorage table. Zero hooks implemented at all currently.
T0.3 — Face-conflict diameter gate	Separate phase	Guard rail for Track 2 T2.1. Should land immediately before Track 2 kickoff.
B35A/B25A stranded hypothesis check	One read-only check	Does t1_geometry_fusion_summary.json have computed dia/spacing/qty? Determines Track 2 first task.
Pattern C — Set2 B26 mis-association	Separate investigation	GT dia=10, associated text says Y8. RC-3 upstream issue, not a crop problem.

3.3 Medium-term — Track 2 (Vision LLM Arbiter)

Sub-phase	Addresses	Target Impact
T2.1 — Face assignment VLM arbiter	RC-3: 137 WRONG_DIAMETER TOP/BOTTOM_MAIN swaps	WRONG_DIAMETER 176→<40
T2.2 — Extra vs main disambiguation	RC-4: 114 TOP_EXTRA + 40 BOTTOM_EXTRA phantom rows	Extra 154→<25
T2.3 — Orphan stirrup confirmation	Remaining 31 MISSING stirrups (synthesised geometry hypotheses)	MISSING 31→<10

Note: Model selection still required: test claude-sonnet-4-6 vs claude-haiku-4-5 on 10 hand-picked WRONG_DIAMETER beams before committing to Track 2 at full scale.

3.4 Still Fully Open

Item	Why Open
T0.2b hook generation at scale	Blocked on Track 1 completion — needs parent stirrups reliable
Spacer residual 33→~0	Blocked on Track 2 extra-bar recovery supplying MAIN+EXTRA pairs
Net Y25 diff% <15%	Currently 27–42% net-of-ACCEPTABLE_EXTRA; same upstream dependency
Per-set overall accuracy ≥80/80/72%	Unknown until Track 2 lands — blended 70.23% not per-set
Bar match ≥70%	At 27.41%, 42+ points short — entire point of Track 2
Bar detection ≥85%	At 68.52%, 16.5 points short — residual stirrups + main-bar recovery

4. Blockers

4.1 Active Blockers (Right Now)

BLOCKER 1 — Y-axis Crop Extent Structural Gap (Pattern A, 40% of Residual)

The crop's vertical extent is derived entirely from R.1 text-anchor Y-positions. For 46 of 115 residual beams, R.1's text anchors don't reach the stirrup tick-row Y-band (typically 1,500–2,500mm away from where the anchor sits). The scaled diagnostic confirmed this pattern across all 3 sets (Set1 46%, Set2 50%, Set3 28%) with the majority of missing bands (29/46) sitting above the beam-mark label rather than below it.

The fix — deriving the vertical extent from geometric signals (-S-STIRUP DIMENSION bbox, beam outline horizontals, section depth window near the mark) independently of R.1 anchor coverage — is approved and ready for implementation. Until this lands and passes the visual gate, Part D (the 115-beam full re-run) cannot execute, and it remains unknown how much of the 31 MISSING / 113 WRONG_QTY residual is recoverable without Track 2.

BLOCKER 2 — Visual Gate Cannot Pass on Text/Label Alone

Part C (content-based 4-check gate) is explicitly blocked until B9/B10 crops visually show stirrup tick geometry matching their manual references. B9 currently shows correct text but no tick marks — a crop useless for detection even if it would pass a naive label-presence check. This is a direct dependency of BLOCKER 1.

4.2 Known Technical Debt (Non-blocking, Should Be Addressed)

Item	Risk Level	Owner / Timing
R.1 runner was loading V8 source for all prior V9 runs	Low (V8/V9 were byte-identical pre-9.2.0)	Monitor each phase
OpenCV 0.45 confidence < 0.55 fusion bar — real detections blocked	Medium (7 beams permanently blocked until reviewed)	Track 2 review
QA.2A SYNTHESIZED_GEOMETRY column still 0	Low (2 beams only)	Track 2 first task
B35A/B25A unverified stranded hypothesis	Low — may already have computed values	Read-only check
B26 (Set2) Pattern C mis-association (GT dia=10, assoc dia=8)	Medium — RC-3 production impact	Track 2 / T0.3
TYPICAL STIRRUP DETAILS shared callout block could contaminate crops	Low — only if Y-extent expands near that sheet region	Handle in B8 legibility fix

5. Next Steps — Ordered by Dependency

Immediate — Complete 9.3.4

Step 1 — Implement geometric Y-extent floor (Part B-2 Step 2)

Fix beam_extent.py: use -S-STIRUP DIMENSION Y-span + beam outline horizontals + section depth window near the mark to set the vertical crop extent — NOT R.1 text-anchor Y-coverage. Apply the 29-above/12-below asymmetry finding: extend further above the mark by default since upper bands are more commonly missed. Derive the actual extent from geometric signals rather than hardcoding direction. Do NOT simply add more padding — demonstrated to cause B8/B13 bleed in earlier iterations.

Step 2 — Pass Part C visual gate

For B1, B2, B8, B9, B10: regenerated crops must show stirrup tick geometry matching manual references, not just label/text presence. B8 multi-zone (two separated tick bands with whitespace between) should be documented as a known-valid visual pattern. The 4-check gate must pass: (1) label completeness, (2) single-beam-only content, (3) no edge truncation, (4) ink density.

Step 3 — Part D full 115-beam re-run

Re-run T1.2 detection with fixed crops, re-apply content-based gate, full 3-set QA.2A. Report honestly whether the residual (31 MISSING / 113 WRONG_QTY) shrinks meaningfully. Either answer is informative — if no change, that now constitutes solid evidence (not an environment-gap assumption) that Track 2 is genuinely the next required investment.

Short-term — After 9.3.4

Step 4 — B35A/B25A stranded hypothesis check (read-only, ~30 min)

Does t1_geometry_fusion_summary.json contain computed dia/spacing/qty for these 2 beams? Determines whether Track 2 starts with a small materialisation task or genuinely fresh synthesis.

Step 5 — T0.3 face-conflict gate

Prerequisite for Track 2 T2.1. Define the FACE_CONFLICT trigger condition (same annotation claimed by both TOP and BOTTOM faces, or one face empty while the other has 2 candidates), implement behind a config flag, confirm it fires on known RC-3 beams.

Step 6 — T0.2a stirrup hook generation

For the 27+ beams whose stirrups now reliably reach production (post-9.2.0/9.3.x). Hook length from R.2A anchorage table. Zero hooks currently implemented. No Track 2 dependency.

Step 7 — Track 2 model selection test

10 hand-picked WRONG_DIAMETER beams, run claude-sonnet-4-6 vs claude-haiku-4-5 face-assignment prompt, measure agreement with estimator GT. Pick winner before scaling to full T2.1.

Medium-term — Track 2 Proper

Step 8 — T2.1 Face Assignment VLM (RC-3)

Narrow structured question per FACE_CONFLICT beam (TOP/BOTTOM/SIDE/SECTION JSON). JSON-schema validated, one retry, deterministic offline fallback. Config flag enable_vision_arbiter. Expected to address RC-3's 137 WRONG_DIAMETER rows. Every VLM verdict logged with tile image hash into vision_evidence.json for auditability.

Step 9 — T2.2 Extra/Main Disambiguation (RC-4)

Full-span vs curtailed-extra + cross-view duplicate detection. Addresses RC-4's 114 phantom TOP_EXTRA rows. Same JSON-schema validation discipline as T2.1.

Step 10 — T2.3 Orphan Stirrup Confirmation

Confirm B35A/B25A and remaining 31 MISSING synthesised hypotheses. Answers: 'Does this section view show a closed stirrup? How many legs?'

Step 11 — Track 3 Gate Check: per-set targets

Run QA.2A on all 3 sets post-Track-2 and measure per-set overall accuracy against $\geq 80\%$ / $\geq 80\%$ / $\geq 72\%$ targets. Check bar detection $\geq 85\%$, bar match $\geq 70\%$, steel KG $\geq 92\%$ simultaneously. This is the program's definition-of-done gate for V9.0.0.

6. Architecture Principles & Standing Conventions

6.1 Core Architecture Principle

The deterministic V8 spine stays **authoritative and reproducible**. OpenCV and Vision LLM enter as **evidence providers** feeding Phase R.2.1D (Evidence & Hypothesis Engine) and R.3.1 (Drawing Relationship Engine) — never as direct mutators of `beam_reinforcement_models_production.json`. All new stages are additive runners under `Version9/Run_PY/` with their own `src/PhaseV9_*` packages, run-context aware, with soft-exit semantics matching the existing spine.

6.2 Safety Discipline — Required for Every Phase

- Never modify frozen Version8; all work in Version9.
- New logic behind config flags; purely additive to production model; soft-exit + artefact-presence semantics.
- Every change followed by QA.2A regression on all 3 benchmark sets; compare role×status matrix to baseline.
- Scope-leakage proof (byte/row-identical output on out-of-scope beams) is the highest-priority regression check.
- Flag-off equivalence — with config flag false, output must be row-identical to the previous baseline.
- Read-only diagnostics must not touch pipeline code or artefacts.
- Every prompt includes: risk section with mandatory fallback, QA carve-out amendment, flag-off equivalence as a hard requirement.
- VLM verdicts logged (tile hash + JSON) and disagreements feed R.1.5/R.1.6 so the rule engine absorbs VLM judgements over time.

6.3 Root Cause Framework

#	Root Cause	Evidence	Fix Class	Status
RC -1	Stirrups & hooks not reaching output	182 STIRRUP + 81 HOOK missing; Y10 98% missing	Discovery + association + piece generation	Partly addressed (9.2.0+9.3.x); hooks open
RC -2	Spacer bars never computed	98/98 SPACER_BAR missing; Y25 82-100% missing	Pure rule computation — no CV needed	Partly done (33 residual; upstream dependency)
RC -3	Main-bar diameter swaps (top/bottom)	137/176 WRONG_DIA are TOP/BOTTOM_MAIN with correct qty	Spatial association — VLM face arbiter	Not started — Track 2 T2.1
RC -4	TOP_EXTRA/BOTTOM_EXTRA over-generation	114+40 EXTRA rows with est_qty=0	Deduplication + role placement — VLM	Not started — Track 2 T2.2

6.4 Prompt Generation Conventions

- Prompts always delivered as a single code block for direct Cursor paste.
- Every prompt includes: risk section, mandatory fallback behaviour, QA carve-out amendment, explicit DO NOT IMPLEMENT section.
- Every phase requires flag-off equivalence proof, scope-leakage proof, and root-cause attribution before commits are accepted.
- Diagnostic phases (read-only scans, root-cause attribution) precede implementation phases.
- Acceptance criteria are specific, measurable, and agreed before implementation begins.
- When Cursor's output is received, it is independently validated here (Claude chat) before commit is approved.
- Cursor sometimes hallucinates — always verify output against attached raw artefacts, not just prose summaries.

7. Summary Assessment

7.1 Overall Programme Health

Steel KG Accuracy	91.07%	≥92%	Essentially at target. The DIMENSION-channel patch (9.2.0) delivered this almost entirely on its own.
Bar Detection	68.52%	≥85%	16.5 points to close. Residual stirrups (31 MISSING) + main-bar recovery via Track 2 should close most of this.
Bar Match	27.41%	≥70%	Largest gap at 42+ points. This is the metric Track 2 (face assignment + extra dedup) exists to move.
Overall Accuracy	70.23%	≥80/80/72%	Per-set breakdown not yet confirmed. Blended figure is 70.23% — about 10 points short on a blended basis.

7.2 Closing Statement

The programme is **ahead on KG accuracy** (91% vs ≥92% target, essentially there), **on track on bar detection** (68% vs 85% target, ~17 points to close with Track 2), and **significantly behind on bar match** (27% vs 70% target, the metric Track 2 exists to move). The crop/render infrastructure work was unplanned and has consumed more iteration than any algorithm work — but the scaled diagnostic (Pattern A: 40% of residual) has now given a clear, evidence-backed pivot direction (geometric Y-extent floor), and once Part C's visual gate passes, the path to Track 2 is clear.

The fundamental architecture decisions — deterministic spine + additive evidence providers, config-flagged changes, flag-off safety proofs, scope-leakage checks — have held throughout and remain sound. The phased diagnostic-first discipline has paid dividends twice: first when the Day-1 stirrup diagnostic revealed a cheap text fix where an expensive geometry engine was planned, and second when the scaled Y-band coverage check prevented unlimited per-beam Y-axis surgery by confirming the gap was systemic rather than incidental.